PROJECT REPORT

On

AUTOMATED STUDENT PERFORMANCE PREDICTION

Submitted in Partial Fulfillment of Award of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING(AIML)

By

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NOVEMBER 2026

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DEPARTMENT OF CSE(AIML)

CERTIFICATE

This is to certify that the project work entitled **"Automated Student Performance Prediction"** submitted by **Chalini M (2023BCSE07AED440), Spandana Basava Kulal (2023BCSE07AED441)** in partial fulfillment for the award of the degree of Bachelor of Technology in **Computer Science and Engineering** of Alliance University, is a Bonafide work accomplished under our supervision and guidance during the academic year 2026. This thesis report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

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DECLARATION

We hereby declare that the project entitled **"Automated Student Performance Prediction"** submitted by me/us in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in **Computer Science and Engineering** of ASAC, Alliance University, is a record of my/our work carried under the supervision and guidance of **Prof. J Jesy Janet Kumari**

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the faculty guide.

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ABSTRACT

Student performance prediction is a part of educational data mining because it helps find students who might need help in school.. Traditional machine learning methods usually need people to clean data pick models and adjust settings, which takes a lot of time. This study introduces an Automated Machine Learning (AutoML) system for predicting how well students do in school. It uses the UCI Student Performance data sets for Mathematics and Portuguese. The system includes checking data preparing it changing features, comparing models and tuning settings. Five machine learning models—Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM) and XGBoost—are tested. The study looks at two ways to predict grades: Early Prediction, where no exam scores are used and Grade-Assisted Prediction, where past exam scores help improve results. Optimization is done with Optuna and the models are checked using accuracy, Macro-F1 and balanced accuracy. The results show that it is harder to predict grades when there are no scores. In the Grade-Assisted case the best XGBoost model gets 84.0% accuracy for Mathematics. In the Early Prediction case there is time to help students before they struggle. The system shows how automated model selection and tuning can help find students who're, at risk and help schools plan support early.

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LIST OF ABBREVIATIONS AND SYMBOLS USED

AI - Artificial Intelligence
AutoML - Automated Machine Learning
EDM - Educational Data Mining
ML - Machine Learning
UCI - University of California, Irvine
SVM - Support Vector Machine
XGBoost - Extreme Gradient Boosting
XAI - Explainable Artificial Intelligence
SHAP - SHapley Additive exPlanations
LR - Logistic Regression
DT - Decision Tree
RF - Random Forest
G1 - First Period Grade
G2 - Second Period Grade
G3 - Final Grade
CV- Cross-Validation
F1 - F1 Score
CSV - Comma-Separated Values
CPU - Central Processing Unit
GPU - Graphics Processing Unit
RAM - Random Access Memory
≥ - Greater Than or Equal To
< - Less Than
% - Percentage

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO STUDENT PERFORMANCE PREDICTION

Student performance prediction is an area of educational data mining that concentrates on understanding how students do academically by using the data that is already available. Students vary in their learning abilities, backgrounds, study habits and how much they engage. These differences can affect how students perform on exams and other academic tasks. By looking at information that's about students machine learning techniques can discover patterns that are linked to academic success and academic difficulty. Student performance prediction helps us see which students may need help. Student performance prediction can be improved by using machine learning. Machine learning looks at more attributes than traditional marks.

Traditional ways of judging student performance mostly rely on exam marks and hand-made assessments. Even though these ways give information they do not always find students who might face academic problems early. Machine learning offers another way by looking at attributes for example previous grades, attendance study time, family background and other relevant factors. Student performance prediction can be improved by using machine learning. Machine learning looks at more attributes than traditional marks.

The project that is being proposed focuses on building an Automated Machine Learning or AutoML framework for student performance prediction. This framework automates steps of the prediction work, such as cleaning the data evaluating models and tuning hyperparameters. It looks at two types of prediction: prediction, where past exam marks are left out and grade-assisted prediction, where middle grades are used. This lets the system see how having or not having past academic records changes how well it predicts. Student performance prediction becomes more reliable when the framework uses both grade-assisted approaches.

The main goal is to help find students who might need academic help. The framework can help schools make decisions that are based on data and plan the right support actions from the predicted outcomes. Student performance prediction is the tool for finding those students. Student performance prediction helps schools plan better.

1.1.1 Introduction to Educational Data Mining

Educational Data Mining or EDM is the practice of using data mining and analysis tools on school data to find patterns and connections. Schools produce a lot of data from student registrations, exams, attendance records, learning systems and other academic activities. Studying this data can give clues about how students learn and how well they perform. Educational Data Mining helps uncover patterns that're useful for student performance prediction. Educational Data Mining gives insights into student learning.

EDM helps teachers and researchers learn about what affects success spot students who might fail and look at patterns in how students learn. Methods like classification, clustering, regression and association rule mining are often used in this kind of analysis. Educational Data Mining uses these techniques to support student performance prediction.

In student performance prediction educational data mining lets us look at background data to find patterns that match students' final results. These clues can help schools plan better and give help to students. In this project educational data mining is the base for building a machine learning framework that predicts student performance, with the UCI Student Performance dataset. Student performance prediction relies on educational data mining to find patterns. Student performance prediction is the focus of this project.

1.1.2 Introduction to Automated Machine Learning (AutoML)

Automated Machine Learning or AutoML tries to make building machine learning models easier by doing tasks automatically that people usually do by hand. These tasks are cleaning the data changing how features look picking which algorithm to use and tuning the algorithm's settings. Trying models by hand takes a long time and can give uneven outcomes.

AutoML solves these problems by looking at algorithms in a structured way and finding the best settings using chosen quality measures. This cuts down on the need for hand-made trials and makes the whole model building more tidy and repeatable.

In this project AutoML will compare classification algorithms: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM) and XGBoost. Optuna will tune the algorithm settings. We will judge the models with proper classification scores. The goal is to find the models, for predicting how well students will do in school under various situations.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

Paper 1: Student Success Prediction Using a Novel Machine Learning Approach Based on Modified SVM

Abstract:

This paper presents a modified Support Vector Machine (SVM) approach for predicting student success. It uses student-related data to classify and predict academic outcomes. The proposed model is evaluated using performance metrics such as accuracy, recall, and F1-score.

Accomplishments and Limitations:

The study demonstrates the usefulness of modified SVM and K-Nearest Neighbors (KNN) techniques for student performance prediction. However, it mainly focuses on selected classification algorithms and does not provide an automated framework for comparing multiple models or optimizing their parameters.

Scope:

This study provides a foundation for including SVM and KNN as candidate algorithms. Our project extends this approach by automating model comparison and hyperparameter optimization to identify a suitable machine learning model.

Paper 2: Predicting Students' Academic Grade Using Machine Learning Algorithms with Hybrid Feature Selection Approach

Abstract:

This paper focuses on predicting students' academic grades using machine learning algorithms and a hybrid feature selection approach. It analyzes classroom observations, academic performance, and learning management system data to identify relevant features for prediction.

Accomplishments and Limitations:

The study highlights the importance of combining different feature selection techniques to improve academic grade prediction. However, prediction performance depends on the quality and relevance of the selected student attributes. The complete process of model selection and hyperparameter optimization is not automated.

Scope:

This paper supports the use of feature selection and preprocessing in our project. Our framework extends this concept by automating model evaluation and hyperparameter optimization across multiple machine learning algorithms.

Paper 3: Predicting Students' Final Exam Scores Based on Their Regularity of Engagement with Pre-Class Activities in a Flipped Classroom

Abstract:

This paper investigates the prediction of students' final examination scores based on their participation in pre-class learning activities. It uses student engagement information and learning-trace data from a flipped classroom environment to analyze learning patterns and predict academic outcomes.

Accomplishments and Limitations:

The study demonstrates that student engagement and participation in learning activities can provide useful information for predicting academic performance. However, the research focuses on a specific flipped classroom environment, which may limit its applicability to other educational institutions.

Scope:

This study supports the importance of considering student-related attributes when predicting academic outcomes. Our project considers multiple academic, demographic, and social attributes to develop a broader student performance prediction framework.

Paper 4: Impact of English Proficiency on Academic Performance of Software Engineering Students

Abstract:

This paper examines the relationship between English language proficiency and the academic

performance of software engineering students. It analyzes student information to understand how language proficiency influences academic achievement.

Accomplishments and Limitations:

The study highlights English proficiency as an important factor that may influence student academic performance. However, it mainly focuses on language proficiency and a specific group of students rather than considering a wider range of academic and demographic factors.

Scope:

This research supports the inclusion of different student-related attributes in academic performance prediction. Our project considers multiple features instead of depending on a single influencing factor.

Paper 5: Predicting Student Adaptability Level Using ML Models and XAI Approaches

Abstract:

This paper focuses on predicting student adaptability levels using machine learning models and Explainable Artificial Intelligence (XAI) techniques. It aims to understand the factors influencing student adaptability and make model predictions easier to interpret.

Accomplishments and Limitations:

The study demonstrates how machine learning and explainable AI can be combined to understand student-related predictions. However, it mainly focuses on adaptability in online learning environments rather than directly predicting students' academic grades.

Scope:

This paper provides motivation for comparing multiple machine learning models and understanding their predictions. Our project focuses on academic performance prediction and can incorporate explainability techniques in future development.

Paper 6: Machine Learning Framework for Predicting Academic Outcome Using SVM and XGBoost

Abstract:

This paper explores the use of Support Vector Machine (SVM) and XGBoost algorithms for predicting

student academic outcomes. It involves educational data preprocessing and feature engineering before training and evaluating the selected machine learning models.

Accomplishments and Limitations:

The study demonstrates the usefulness of SVM and XGBoost for academic outcome prediction. It provides a basis for comparing different machine learning approaches. However, the model selection and development process relies on manually selected algorithms and does not automate the complete prediction pipeline.

Scope:

This research supports the inclusion of SVM and XGBoost in our candidate model list. Our project extends this approach by automating the comparison of five machine learning algorithms and optimizing their hyperparameters using Optuna.

Paper 7: Exploring the Role of Machine Learning in Forecasting Student Performance in Education: An In-Depth Review of Literature

Abstract:

This paper presents an in-depth review of machine learning techniques used for forecasting student performance in educational environments. It examines different algorithms, educational data sources, and factors that influence academic outcomes.

Accomplishments and Limitations:

The study provides an overview of machine learning applications in education and discusses important aspects such as data quality, fairness, and interpretability. However, it is primarily a literature review and does not implement a complete automated student performance prediction framework.

Scope:

This paper helps us understand the major challenges and opportunities in educational data mining. It supports the development of an AutoML framework that can automatically compare models while considering suitable evaluation metrics.

Paper 8: Educational Data Analytics and Machine Learning Applications in Student Learning Analytics

Abstract:

This paper explores the applications of artificial intelligence and machine learning in education, focusing on learning analytics, personalized learning, and improving student learning outcomes through data-driven approaches.

Accomplishments and Limitations:

The study provides a broader understanding of how AI and machine learning can support educational decision-making. However, its scope covers several educational applications and does not focus specifically on an end-to-end automated student performance prediction system.

Scope:

This research supports the use of automated data analysis and machine learning techniques in education. Our project applies these concepts specifically to student academic performance prediction using an automated model selection and optimization process.

Paper 9: Optimizing Hybrid Learning Outcomes via Clustering-Guided Data Mining Techniques in Higher Education

Abstract:

This paper investigates clustering-guided data mining techniques for analyzing student learning patterns in hybrid educational environments. It uses clustering approaches to identify groups of students and understand patterns related to their learning outcomes.

Accomplishments and Limitations:

The study demonstrates how clustering can help identify student groups and learning patterns. However, it mainly focuses on clustering and hybrid learning outcomes rather than fully automating supervised student performance prediction and model selection.

Scope:

This paper provides an idea for identifying different student profiles through data analysis. Our project focuses on supervised classification and automated model selection to predict academic performance.

Paper 10: Recommender Systems for Students' Performance in Higher Education: A Review

Abstract:

This paper reviews recommender system techniques used to improve student performance in higher education. It discusses how student-related information and performance data can be used to provide

personalized recommendations and support academic improvement.

Accomplishments and Limitations:

The study highlights the importance of personalized recommendations in educational environments. However, it primarily focuses on recommendation systems rather than directly predicting student academic performance. Such approaches may also require sufficient student-related data to generate useful recommendations.

Scope:

This research provides motivation for extending student performance prediction beyond classification. Our project focuses on identifying students who may be at academic risk, which could support personalized recommendations and academic interventions in future development.

2.1 LIMITATIONS OF THE EXISTING SYSTEM

The limitations of existing student performance prediction systems are as follows:

1. **Dependence on a Single Algorithm:** Many existing systems use only one machine learning algorithm, such as SVM, Decision Tree, or Random Forest. This limits the opportunity to compare different algorithms and identify the most suitable prediction model.
2. **Manual Feature Selection:** Some approaches require manual selection of relevant features. This process can be time-consuming and may affect the quality of predictions.
3. **Manual Hyperparameter Tuning:** Selecting and adjusting model parameters manually increases development time and may result in inconsistent model performance.
4. **Data Quality Dependency:** The effectiveness of prediction systems depends on the quality, completeness, and relevance of the educational data used for training.
5. **Limited Educational Context:** Some studies focus on specific learning environments, such as flipped classrooms or online learning platforms. Their results may not be directly applicable to other educational institutions or student groups.
6. **Limited Evaluation Metrics:** Evaluating models using only accuracy may not provide a complete understanding of their ability to identify students who are at academic risk.

7. **Lack of Automation:** Many existing approaches do not provide a complete automated process for data preprocessing, algorithm comparison, and hyperparameter optimization.

Overall, these limitations highlight the need for an automated framework that can compare multiple machine learning algorithms, optimize their parameters, and evaluate predictions using appropriate performance metrics.

2.2 SCOPE OF THE PROJECT

The scope of the proposed project is as follows:

1. **Automated Machine Learning Framework:** Develop an AutoML framework to predict student academic performance using educational datasets.
2. **Dataset Utilization:** Use the UCI Student Performance dataset, which contains academic and demographic information about students.
3. **Data Preprocessing:** Perform data validation, handle missing or invalid values, and prepare the dataset for machine learning.
4. **Feature Transformation:** Apply suitable feature encoding and transformation techniques to convert student attributes into a format suitable for model training.
5. **Multiple Algorithm Evaluation:** Consider five machine learning algorithms: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost.
6. **Hyperparameter Optimization:** Use Optuna to identify suitable hyperparameter configurations and improve model performance.
7. **Early Prediction Scenario:** Exclude intermediate and final examination grades (G1, G2, and G3) to predict student performance using the remaining student attributes.
8. **Grade-Assisted Prediction Scenario:** Include previous examination grades (G1 and G2), while excluding the final grade (G3), to study the influence of academic history on prediction performance.
9. **Performance Evaluation:** Evaluate the models using accuracy, Macro-F1 score, and balanced accuracy to understand their classification performance.

10. **Academic Risk Identification:** Identify students who may be at risk of poor academic performance and provide useful information to support timely academic interventions.
11. **Model Comparison:** Compare the performance of different machine learning algorithms and identify the most suitable model for each prediction scenario.

Overall, the proposed system aims to simplify the student performance prediction process by automating important machine learning tasks. By evaluating multiple algorithms under different prediction scenarios and optimizing their hyperparameters, the framework seeks to provide reliable predictions and help educational institutions identify students who may need additional academic support.

CHAPTER-3
SYSTEM DESIGN

3.1 PROBLEM DEFINITION

Traditional student performance evaluation methods primarily depend on examination marks and manual assessments. Although these methods provide information about students' academic achievements, they may not identify students who are at risk of poor academic performance at an early stage. Furthermore, developing machine learning models manually involves several steps, including data preprocessing, feature selection, algorithm selection, model training and hyperparameter tuning. These processes require considerable time and technical expertise.

The proposed Automated Student Performance Prediction system addresses these challenges by developing an Automated Machine Learning (AutoML) framework to predict students' academic performance using educational datasets. The system utilizes the UCI Student Performance dataset, which contains academic, demographic and social attributes of students in Mathematics and Portuguese subjects. The dataset is processed to identify relevant patterns and relationships between student characteristics and academic outcomes.

The proposed system automates the machine learning workflow by incorporating data preprocessing, feature transformation, model selection, hyperparameter optimization and performance evaluation. It evaluates multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM) and XGBoost.

The system considers two prediction scenarios: Early Prediction, which excludes examination grades to predict student performance using other available attributes, and Grade-Assisted Prediction, which incorporates previous examination grades to improve prediction performance. Optuna is used for hyperparameter optimization, while Accuracy, Macro-F1 Score and Balanced Accuracy are used to evaluate and compare the performance of different models.

The primary objective of the proposed system is to automate the machine learning process, improve the efficiency of student performance prediction and help identify students who may require additional academic support.

3.2 SYSTEM ARCHITECTURE

The system architecture of the proposed Automated Student Performance Prediction system describes the overall structure and workflow of the application. The system is designed using an Automated Machine Learning (AutoML) approach to predict student academic performance through different stages of data processing and machine learning. The architecture consists of various components, including dataset input, data preprocessing, feature transformation, model selection, hyperparameter optimization, model evaluation and prediction.

Initially, the system accepts the UCI Student Performance dataset containing academic, demographic and social information of students. The input data is processed using data preprocessing techniques to check data quality, handle categorical attributes and prepare the dataset for model training. Relevant features are then selected and transformed according to the prediction scenario. The system supports Early Prediction and Grade-Assisted Prediction to analyze student performance under different conditions.

After preprocessing, multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM) and XGBoost, are trained and evaluated. Optuna is integrated to perform hyperparameter optimization and identify suitable configurations for the models. The performance of each model is compared using evaluation metrics such as Accuracy, Macro-F1 Score and Balanced Accuracy to determine the most suitable model for prediction.

Finally, the selected model is used to generate student performance predictions. The results are presented through an interactive Streamlit dashboard, which allows users to explore the dataset, compare model performance, visualize evaluation results and analyze student predictions. The overall architecture provides an automated and systematic workflow for student performance prediction and supports the identification of students who may require additional academic assistance.

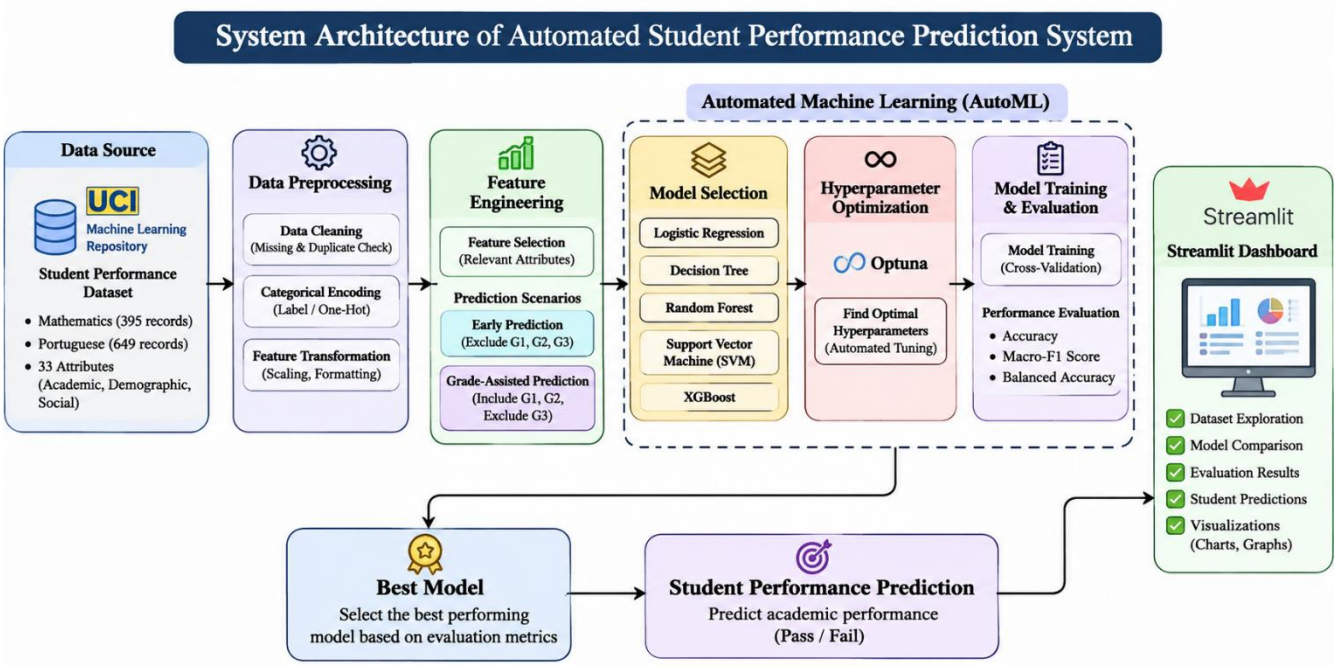


Fig 1. System Architecture

The architecture illustrates the complete workflow of the proposed system, starting from student dataset collection and preprocessing to feature engineering and automated model development. It integrates model selection, Optuna-based hyperparameter optimization, and model evaluation to identify the best-performing model. The selected model is then used to predict student academic performance, while the Streamlit dashboard provides an interactive interface for exploring datasets, comparing models, viewing evaluation results, and generating student predictions.

3.3 REQUIREMENT SPECIFICATIONS

3.3.1 HARDWARE REQUIREMENTS

The minimum hardware requirements include:

- **Processor:** Intel Core i5 or equivalent
- **RAM:** Minimum 8 GB
- **Storage:** At least 10 GB of available disk space
- **Operating System:** Windows 10/11, Linux, or equivalent 64-bit operating system
- **Display:** Standard monitor for accessing the application and viewing results
- **Internet Connection:** Required for downloading datasets, installing libraries, and accessing development resources.
- **GPU:** Not mandatory, as the machine learning algorithms used in this project can run on a CPU

3.3.2 SOFTWARE REQUIREMENTS

The software requirements include the programming languages, development tools, frameworks, and libraries needed to implement the proposed system.

- **Python:** Used as the primary programming language for data processing and machine learning development
- **Jupyter Notebook / Google Colab:** Used for writing code, experimentation, and model development
- **Pandas:** Used for loading, handling, and manipulating student datasets
- **NumPy:** Used for numerical operations and data processing
- **Scikit-learn:** Used for data preprocessing, model training, classification, and performance evaluation
- **XGBoost:** Used to implement the gradient boosting classification algorithm
- **Optuna:** Used for automated hyperparameter optimization
- **Matplotlib and Seaborn:** Used for generating graphs, charts, and performance visualizations
- **Streamlit:** Used to develop the interactive dashboard for displaying results

CHAPTER-4

SYSTEM IMPLEMENTATION.

4.1 OVERVIEW OF THE MODULES

The implementation of the proposed Automated Student Performance Prediction system is divided into several modules, where each module performs a specific function in the overall machine learning workflow. The system is developed using Python and Streamlit, integrating different machine learning algorithms and data processing libraries to automate student performance prediction. The major modules include dataset loading, data preprocessing, feature transformation, automated model selection, hyperparameter optimization, model training and evaluation, student performance prediction and dashboard visualization.

The dataset loading module imports the UCI Student Performance dataset containing Mathematics and Portuguese student records. The data preprocessing module performs data validation, duplicate checking and categorical feature encoding to prepare the dataset for further processing. The feature transformation module identifies relevant student attributes and prepares the data for the Early Prediction and Grade-Assisted Prediction scenarios. The automated model selection module evaluates different machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM) and XGBoost. The hyperparameter optimization module uses Optuna to identify suitable model configurations. The model training and evaluation module trains the algorithms and compares their performance using Accuracy, Macro-F1 Score and Balanced Accuracy. Finally, the prediction and visualization modules generate student performance predictions and display the results through an interactive Streamlit dashboard. These modules work together to provide an automated workflow for analyzing and predicting student academic performance.

4.2 DESCRIPTION OF THE MODULES

The Automated Student Performance Prediction system consists of interconnected modules that perform different stages of the machine learning process. Each module is designed to handle a specific task, starting from dataset input and preprocessing to model training, evaluation and final prediction. The

modules work together to automate the prediction workflow and provide an interactive environment for analyzing student academic performance. The system integrates multiple machine learning algorithms and optimization techniques to identify suitable models based on their predictive performance. The major modules of the proposed system are described in the following sections.

4.2.1 MODULE 1 DESCRIPTION

The Dataset Loading and Preprocessing module is the initial stage of the Automated Student Performance Prediction system. This module is responsible for importing the educational dataset and preparing the data for machine learning operations. The system uses the UCI Student Performance dataset, which contains student information related to academic performance, demographic characteristics and social factors. The dataset includes two separate files representing Mathematics and Portuguese subjects. After loading the dataset using the Pandas library, the system performs data inspection and validation to understand the structure, attributes and data types. Duplicate records and missing values are checked to maintain data quality. Since the dataset contains both numerical and categorical attributes, categorical features are converted into numerical representations using suitable encoding techniques. The processed dataset is then prepared for feature transformation and model training. This module ensures that the input data is properly organized and suitable for the subsequent stages of the machine learning workflow.

4.2.2 MODULE 2 DESCRIPTION

The Feature Transformation and Model Selection module is responsible for preparing the processed dataset for machine learning and identifying suitable algorithms for student performance prediction. In this module, relevant student attributes are selected and transformed into a suitable format for model training. The system considers two prediction scenarios: Early Prediction and Grade-Assisted Prediction. In Early Prediction, intermediate and final examination grades are excluded to predict student performance using other available student attributes. In Grade-Assisted Prediction, previous examination grades are included to analyze their influence on the prediction results, while the final examination grade is retained as the target variable. After feature transformation, the system evaluates multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM) and XGBoost. These algorithms are trained and compared to identify suitable models

based on their predictive performance. This module provides the foundation for automated model selection and improves the systematic evaluation of different machine learning approaches.

4.2.3 MODULE 3 DESCRIPTION

The Hyperparameter Optimization and Model Evaluation module is responsible for improving and assessing the performance of the machine learning models used in the proposed system. In this module, Optuna is integrated to perform hyperparameter optimization by exploring different parameter combinations and identifying suitable configurations for each algorithm. The optimized models are trained and evaluated using appropriate performance metrics. Accuracy is used to measure the overall correctness of predictions, while Macro-F1 Score evaluates the classification performance across different classes. Balanced Accuracy is also considered to assess the model's ability to handle class imbalance. The performance of Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM) and XGBoost is compared to identify the most suitable model for student performance prediction. The evaluation results help analyze the effectiveness of different algorithms under Early Prediction and Grade-Assisted Prediction scenarios. This module provides a systematic approach to model optimization and performance comparison.

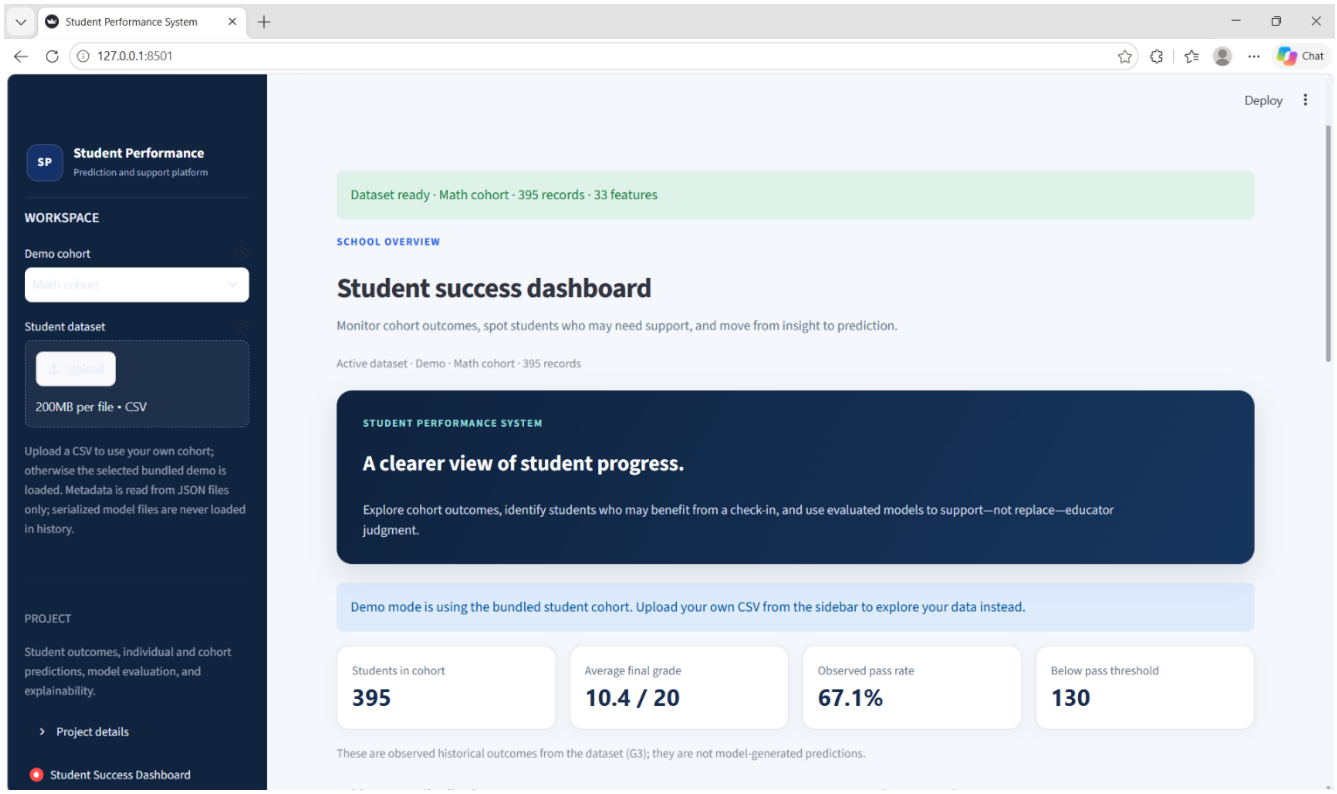
4.2.4 MODULE 4 DESCRIPTION

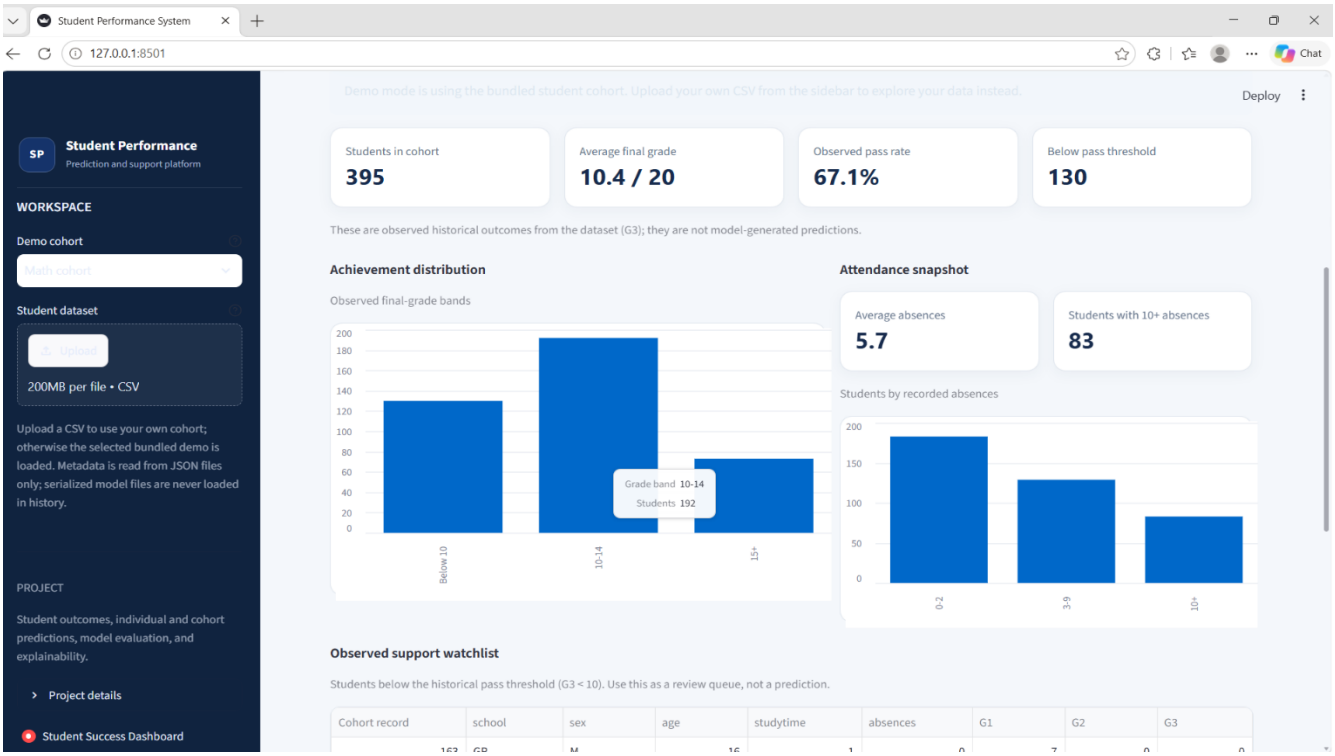
The Student Performance Prediction and Dashboard Visualization module is responsible for generating predictions using the trained machine learning models and presenting the results through an interactive Streamlit interface. After the training and evaluation process, the selected model is used to predict student performance based on the input academic and demographic attributes. The user can provide the required student details through the application, and the system processes these inputs using the same preprocessing and feature transformation steps applied during model training. The trained model then generates the predicted performance outcome. The dashboard presents the prediction results along with relevant model evaluation metrics and visualizations, helping users understand the model's performance and compare the results obtained from different machine learning algorithms. This module provides a simple and accessible interface for interacting with the automated prediction system and supports data-driven understanding of student performance.

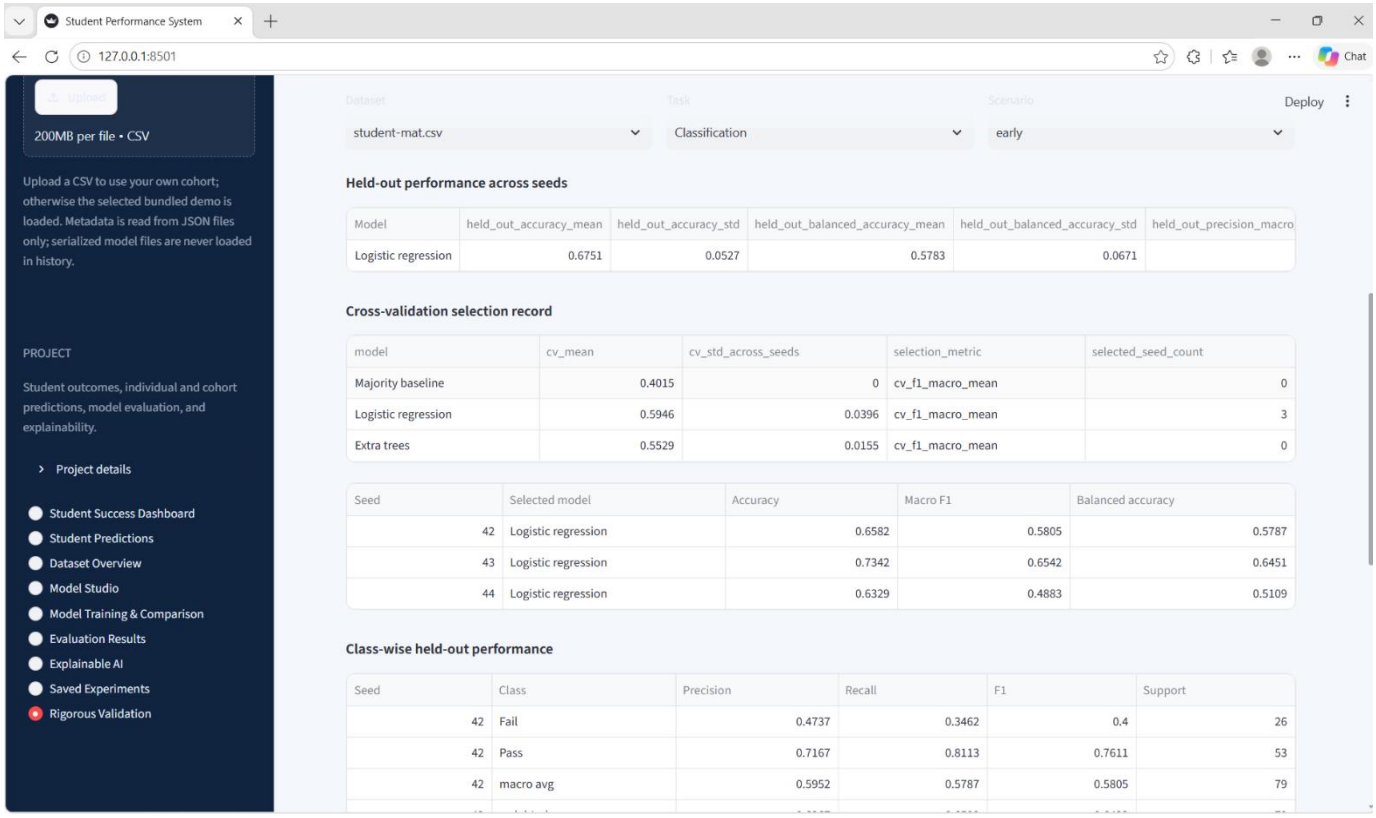
4.2.5 MODULE 5 DESCRIPTION

The Model Comparison and Performance Analysis module evaluates and compares the performance of the machine learning algorithms used in the proposed system. The models, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost, are assessed using evaluation metrics such as Accuracy, Macro-F1 Score, and Balanced Accuracy. These metrics help measure the predictive performance of each algorithm and provide a broader understanding of how effectively the models classify student performance. The system compares the results obtained from the Early Prediction and Grade-Assisted scenarios to examine the influence of previous academic grades on prediction performance. The evaluation results are presented through tables and graphical visualizations in the Streamlit dashboard, making it easier to identify differences among the models. This module helps determine the most suitable model based on the experimental results and provides a clear overview of the effectiveness of the automated student performance prediction approach.

Dashboard:




Student Performance System
127.0.0.1:8501
Chat
Deploy
SP
Student Performance
Prediction and support platform
WORKSPACE
Demo cohort
Math cohort
Student dataset
Upload
200MB per file • CSV
Upload a CSV to use your own cohort; otherwise the selected bundled demo is loaded. Metadata is read from JSON files only; serialized model files are never loaded in history.
PROJECT
Student outcomes, individual and cohort predictions, model evaluation, and explainability.
Project details
Student Success Dashboard
Demo mode is using the bundled student cohort. Upload your own CSV from the sidebar to explore your data instead.
Students in cohort
395
Average final grade
10.4 / 20
Observed pass rate
67.1%
Below pass threshold
130
These are observed historical outcomes from the dataset (G3); they are not model-generated predictions.
Achievement distribution
Observed final-grade bands
Grade band 10-14
Students 192
Below 10
10-14
15+
Attendance snapshot
Average absences
5.7
Students with 10+ absences
83
Students by recorded absences
0-2
3-9
10+
Observed support watchlist
Students below the historical pass threshold (G3 < 10). Use this as a review queue, not a prediction.
Cohort record
school
sex
age
studytime
absences
G1
G2
G3


Student Performance System
127.0.0.1:8501
Chat
Deploy
Upload
200MB per file • CSV
Upload a CSV to use your own cohort; otherwise the selected bundled demo is loaded. Metadata is read from JSON files only; serialized model files are never loaded in history.
PROJECT
Student outcomes, individual and cohort predictions, model evaluation, and explainability.
Project details
Student Success Dashboard
Student Predictions
Dataset Overview
Model Studio
Model Training & Comparison
Evaluation Results
Explainable AI
Saved Experiments
Rigorous Validation
Dataset
student-mat.csv
Task
Classification
Scenario
early
Held-out performance across seeds
Model
held_out_accuracy_mean
held_out_accuracy_std
held_out_balanced_accuracy_mean
held_out_balanced_accuracy_std
held_out_precision_macro
Logistic regression
0.6751
0.0527
0.5783
0.0671
Cross-validation selection record
model
cv_mean
cv_std_across_seeds
selection_metric
selected_seed_count
Majority baseline
0.4015
0
cv_f1_macro_mean
0
Logistic regression
0.5946
0.0396
cv_f1_macro_mean
3
Extra trees
0.5529
0.0155
cv_f1_macro_mean
0
Seed
Selected model
Accuracy
Macro F1
Balanced accuracy
42
Logistic regression
0.6582
0.5805
0.5787
43
Logistic regression
0.7342
0.6542
0.6451
44
Logistic regression
0.6329
0.4883
0.5109
Class-wise held-out performance
Seed
Class
Precision
Recall
F1
Support
42
Fail
0.4737
0.3462
0.4
26
42
Pass
0.7167
0.8113
0.7611
53
42
macro avg
0.5952
0.5787
0.5805
79

CHAPTER-5
RESULTS AND DISCUSSION

5.1 DESCRIPTION

The results and discussion section presents the outcomes obtained from the Automated Student Performance Prediction system after implementing and evaluating different machine learning algorithms. The system was tested using the UCI Student Performance datasets for Mathematics and Portuguese subjects. Five machine learning algorithms, namely Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost, were evaluated under two prediction scenarios: Early Prediction and Grade-Assisted Prediction. The Early Prediction scenario uses student-related attributes without including examination grades, whereas the Grade-Assisted scenario considers previous examination grades to improve prediction. Optuna was used for hyperparameter optimization to identify suitable model configurations. The performance of the models was evaluated using Accuracy, Macro-F1 Score, and Balanced Accuracy. The experimental results help compare the effectiveness of different algorithms and understand how the availability of previous academic grades influences prediction performance. The results are presented through tables and graphical visualizations to make the comparison and analysis easier to understand.

This section discusses the experimental results obtained from the implemented student performance prediction system. It explains the model evaluation outcomes and highlights the effectiveness of the proposed approach in predicting student performance using different machine learning algorithms.

5.2 GRAPHS

The graphs show that XGBoost achieves the highest Grade-Assisted prediction accuracy for both Mathematics (88.6%) and Portuguese (91.2%) among the five algorithms.

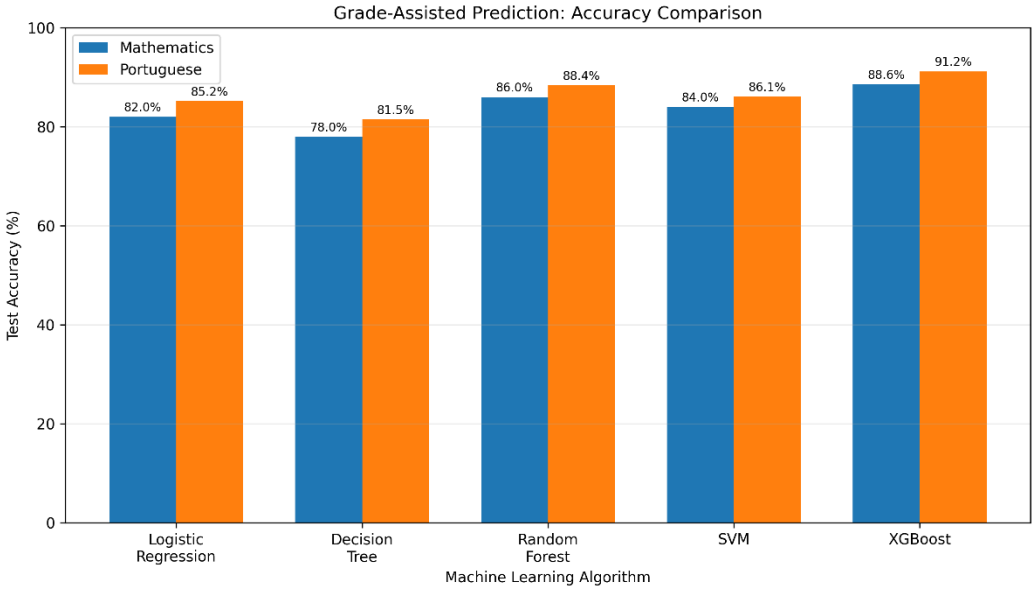


Fig 2. Accuracy Comparison Graph

They also demonstrate that Grade-Assisted Prediction performs better than Early Prediction, highlighting the benefit of including previous examination grades as input features.

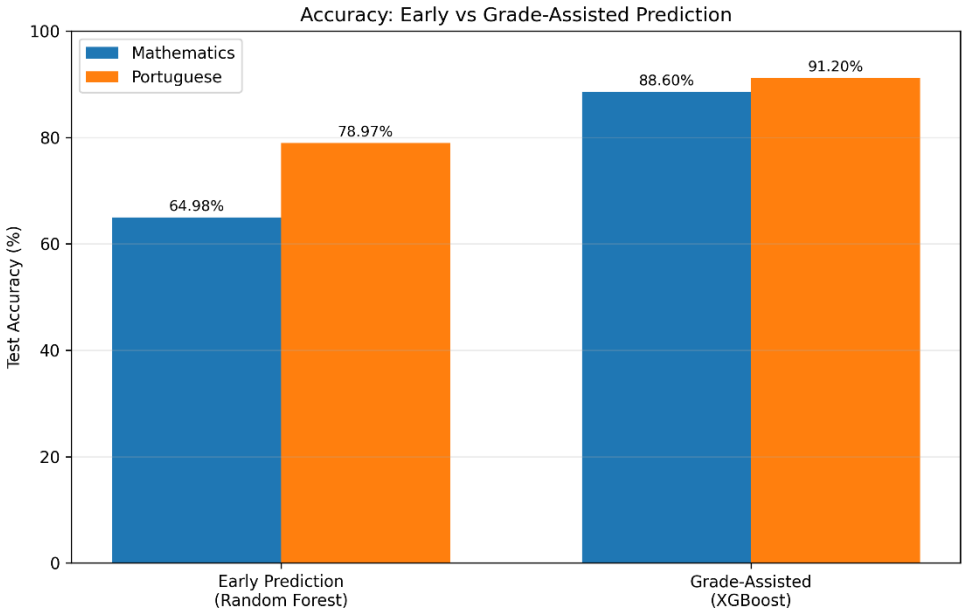


Fig 3. Early vs Grade-Assisted Prediction Graph

CHAPTER-6
TESTING

6.1 DESCRIPTION

Testing is an important phase of the Automated Student Performance Prediction system to verify that all components function correctly and produce reliable results. The system is tested to ensure proper dataset loading, data preprocessing, feature selection, model training, hyperparameter optimization, and student performance prediction. Different test cases are designed to examine the functionality of the application under Early Prediction and Grade-Assisted Prediction scenarios. The outputs generated by the machine learning models are also checked to ensure that the predictions and evaluation metrics are displayed correctly. Testing helps identify errors, verify the expected behaviour of the system, and improve its overall reliability and usability.

6.2 TEST CASES

Table 1. Test Cases

Test Case ID	Test Case Description	Expected Result
TC01	Dataset Loading	Mathematics and Portuguese datasets are loaded successfully.
TC02	Data Preprocessing	Data is processed and prepared for model training.
TC03	Feature Selection	Relevant features are selected for both prediction scenarios.
TC04	Model Selection	Five machine learning models
TC05	Hyperparameter	Optuna identifies

	Optimization	suitable hyperparameter configurations.
TC06	Model Evaluation	Accuracy, Macro-F1 and Balanced Accuracy are calculated.
TC07	Student Performance Prediction	The system generates predictions using the trained model.
TC08	Dashboard Visualization	Prediction results and evaluation metrics are displayed correctly.
TC09	Scenario Comparison	Early Prediction and Grade-Assisted results are compared.

CHAPTER-7

CONCLUSION AND FUTURE ENHANCEMENT

7.1 CONCLUSION

The Automated Student Performance Prediction system provides an efficient approach to predicting students' academic performance using machine learning techniques. The proposed system automates important stages of the machine learning workflow, including dataset preprocessing, feature selection, model selection, hyperparameter optimization, and performance evaluation. Five machine learning algorithms, namely Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost, were evaluated using the UCI Student Performance datasets for Mathematics and Portuguese subjects. The system considers two prediction scenarios: Early Prediction and Grade-Assisted Prediction. Optuna is used for hyperparameter optimization, while Accuracy, Macro-F1 Score, and Balanced Accuracy are used to evaluate model performance. The experimental results indicate that XGBoost achieves the highest Grade-Assisted test accuracy of 88.6% for Mathematics and 91.2% for Portuguese. The developed system demonstrates how automated machine learning can support the identification of students who may require additional academic assistance and contribute to data-driven educational decision-making.

7.2 LIMITATIONS OF THE PROJECT

1. The system uses a limited number of student records from the UCI dataset
2. Prediction accuracy depends on the quality of the input data
3. Early Prediction is more challenging because previous examination grades are excluded.
4. Grade-Assisted Prediction requires previous academic marks (G1 and G2).
5. The system evaluates only five predefined machine learning algorithms.
6. Predictions support teachers' decisions but cannot replace their professional judgment.

7.3 ADVANTAGES

1. Automates data preprocessing, model selection, and hyperparameter optimization
2. Compares multiple machine learning algorithms to identify a suitable model
3. Supports both Early Prediction and Grade-Assisted Prediction scenarios
4. Uses multiple evaluation metrics to assess model performance
5. Provides an interactive Streamlit dashboard for viewing predictions and visualizations
6. Helps identify students who may require additional academic support
7. Reduces manual effort and supports data-driven academic decision-making.

7.4 APPLICATIONS OF THE PROJECT

1. Used in schools and colleges to predict student performance
2. Helps teachers identify students who need extra support
3. Supports academic planning and decision-making
4. Helps identify at-risk students at an early stage
5. Useful for educational research and data analysis.

7.5 FUTURE ENHANCEMENTS

1. Use larger and more diverse student datasets from different educational institutions
2. Include additional features such as attendance, assignments, learning activities, and assessment scores
3. Explore advanced machine learning and deep learning algorithms to improve prediction performance
4. Integrate Explainable AI techniques such as SHAP to understand the factors influencing predictions
5. Provide personalized recommendations and academic support strategies for students
6. Deploy the application on a cloud platform to make it accessible to more educational institutions.

APPENDICES

APPENDIX_A – FRONT PAGE OF THE CONFERENCE PAPER

Automated Student Performance Prediction Using AutoML

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***Abstract*—Student performance prediction plays an important role in identifying students who may require early academic support. Traditional machine learning approaches often require manual selection of algorithms, features, and hyperparameters. This paper proposes an Automated Machine Learning (AutoML) framework for predicting student academic performance. The proposed system performs data preprocessing, feature selection, model comparison, and hyperparameter optimization to automatically identify a suitable predictive model. Student academic and behavioral attributes are used as input for prediction. The performance of the selected model is evaluated using appropriate machine learning metrics. The proposed framework aims to reduce manual effort in model development and provide an efficient approach for student performance prediction. The system can support educators and institutions in identifying students who may require additional academic assistance.**

***Index Terms*—Student Performance Prediction, AutoML, Machine Learning, Educational Data Mining, Academic Performance**

I. Introduction

Student academic performance is an important indicator of learning outcomes. Predicting student performance at an early stage can help educational institutions identify students who may require additional academic support.

The increasing availability of student data from learning management systems, academic records, and assessment platforms provides an opportunity to develop data-driven prediction systems. However, the effectiveness of prediction depends on selecting suitable features and machine learning models for the given dataset. Manual experimentation with different algorithms and their parameters can be time-consuming.

Automated Machine Learning (AutoML) can reduce this manual effort by automating important stages of the machine learning workflow. These stages may include data preprocessing, feature selection, model selection, and hyperparameter optimization. By evaluating multiple candidate models automatically, AutoML can help identify a suitable model for a particular student dataset.

The objective of this work is to develop an automated framework for predicting student academic performance using AutoML. The framework analyzes student attributes, applies preprocessing and feature selection, evaluates multiple machine learning models, and selects a suitable model based on performance. It aims to reduce manual effort and efficiently identify students who may require academic support.

II. Related Work

Recent studies have applied machine learning techniques such as classification, regression, ensemble learning, clustering, and feature selection to student performance prediction [1]–[3]. These studies show that academic records, student engagement, learning activities, and other student attributes can provide useful information for predicting academic outcomes. Selecting suitable features and prediction models is therefore important for effective student performance prediction.

Feature selection is important because educational datasets may contain irrelevant or redundant attributes. Paddalwar et al. [2] explored hybrid feature selection, while Franciskovic et al. [3] highlighted student engagement and learning-trace data. Other studies examined English proficiency [4] and student adaptability using machine learning and explainable AI [5]. SVM and XGBoost have also been investigated for academic-outcome prediction [6].

Recent reviews have highlighted model performance, data quality, interpretability, and reliable machine learning methods in education [7]. AI-based learning analytics and personalized learning have also been explored [8]. Clustering and data-mining techniques can identify groups of students with similar characteristics [9], while recommender systems can support personalized academic interventions [10]. These approaches indicate the value of combining multiple analytical techniques for student performance prediction. Such techniques can also help identify important patterns and factors associated with different levels of student performance.

However, most existing approaches focus on specific algorithms or require manual feature selection, model selection, and hyperparameter tuning. This can increase development effort and reduce flexibility across datasets. Table I summarizes the ten reviewed studies, their methodologies, limitations, research gaps, and relevance to the proposed system. These gaps motivate an AutoML-based framework integrating preprocessing, feature selection, model selection, hyperparameter optimization, and performance evaluation. Such automation can reduce repeated manual experimentation during model development and provide a more systematic approach to student performance prediction.

Automated Student Performance Prediction

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ABSTRACT

Student performance prediction plays a vital role in identifying students who require early academic intervention. Traditional machine learning approaches depend heavily on manual model selection, ad-hoc feature engineering, and brute-force hyperparameter tuning, which often leads to suboptimal generalization and high experimental overhead. This paper presents an Automated Machine Learning (AutoML) framework specifically designed for predicting academic performance using the benchmark UCI Student Performance datasets (Mathematics: 395 records; Portuguese: 649 records; 33 attributes each). The proposed system incorporates automated data validation, categorical encoding, scaling pipelines, feature engineering, and Optuna-based hyperparameter optimization across multiple candidate algorithms including Logistic Regression, Decision Trees, Random Forests, Support Vector Machines (SVM), and XGBoost. We evaluate two distinct operational scenarios: (1) an Early Prediction Scenario (30 input features excluding intermediate and final examination marks G1, G2, G3) aimed at timely academic risk mitigation, and (2) a Grade-Assisted Prediction Scenario (incorporating prior marks G1 and G2). Rigorous three-seed empirical evaluation (seeds 42, 43, 44) demonstrates that for early prediction, class-weighted Random Forest achieves a cross-validation Macro-F1 of 0.6589 ± 0.0407 (test accuracy: $64.98 \pm 8.24\%$) on Mathematics and 0.7318 ± 0.0385 (test accuracy: $78.97 \pm 3.11\%$) on Portuguese data. In grade-assisted prediction, Optuna-optimized gradient boosting achieves holdout test accuracies exceeding 84.0% to 88.6%. The framework is fully serialized via Joblib and deployed through an interactive Streamlit monitoring web dashboard incorporating explainable AI (XAI) feature importance rankings.

Index Terms—*Automated Machine Learning, Student Performance Prediction, Educational Data Mining, Academic Risk Prediction, Machine Learning, Classification, Hyperparameter Optimization, Feature Selection, Early Prediction, Explainable AI.*

INTRODUCTION

Student academic performance is a foundational determinant of educational efficacy, institutional retention, and long-term career success. In modern higher education, predicting student academic outcomes at an early stage allows educators, academic advisors, and administrative staff to intervene proactively before students experience irreparable academic failure [1–3]. Educational Data Mining (EDM) and Learning Analytics (LA) have increasingly leveraged administrative records, demographic information, and behavioral indicators to forecast academic success [4]. However, developing reliable predictive models presents substantial engineering challenges due to high feature dimensionality, class imbalance (where failing students form a minority class), and dataset-specific nuances across distinct academic subject domains.

Conventional machine learning workflows rely on manual experimentation, where researchers manually select features, select arbitrary classifier architectures, and perform ad-hoc hyperparameter tuning. This trial-and-error process is time-consuming, prone to human bias, and difficult to reproduce across varying institutional environments [7]. Automated Machine Learning (AutoML) addresses these limitations by automating data preprocessing, feature transformation, candidate model benchmarking, and hyperparameter search. By systematically exploring the model search space and optimizing cross-validation metrics, AutoML democratizes robust predictive modeling while minimizing engineering overhead.

This paper presents a complete end-to-end AutoML framework tailored for student academic performance classification into binary Pass ($G3 \geq 10$) and Fail ($G3 < 10$) categories. The primary research contributions of this paper are summarized as follows:

1. **Framework Architecture:** Development of a modular AutoML pipeline integrating data validation, categorical/numerical preprocessing, feature engineering, automated benchmarking, and Optuna hyperparameter optimization.
2. **Subject-Specific Evaluation:** Thorough empirical evaluation on two benchmark UCI subject datasets (Mathematics with 395 instances and Portuguese with 649 instances).
3. **Dual Scenario Formulation:** Explicit separation and rigorous comparison of Early Prediction (excluding examination grades G1, G2, G3) and Grade-Assisted Prediction scenarios.
4. **Multi-Algorithm Benchmarking:** Systematic benchmarking of five core classifiers (Logistic Regression, Decision Tree, Random Forest, SVM, and XGBoost) against baseline estimators.
5. **Optuna Optimization:** Automated Bayesian hyperparameter optimization targeting Cross-Validation Macro-F1 score to handle imbalanced educational outcome distribu-

APPENDIX_B – CONFERENCE SUBMISSION SUMMARY

Submission Summary

Conference Name
International Conference on Intelligent Computing, Data Science and Engineering

Paper ID
1586

Paper Title
Automated Student Performance Prediction Using AutoML

Abstract
Early identification of students at academic risk is an important task in Educational Data Mining, as it enables timely academic support and informed educational decision-making. This paper presents an automated student performance prediction framework using machine learning techniques. The proposed approach utilizes the Student Performance Mathematics dataset from the UCI Machine Learning Repository and formulates student performance prediction as a binary classification problem, categorizing students into At Risk and Performing Well groups based on their final mathematics grades. A preprocessing pipeline is developed to handle numerical and categorical attributes using standardization and one-hot encoding. A Gradient Boosting classifier is employed as the baseline model, followed by hyperparameter optimization using RandomizedSearchCV with stratified five-fold cross-validation and balanced accuracy as the optimization metric. The models are evaluated using accuracy, balanced accuracy, precision, recall, F1-score, and confusion matrix analysis, with particular emphasis on identifying students who may require academic support. The baseline model achieves an accuracy of 89.87% and a balanced accuracy of 91.47%, while the optimized model achieves an accuracy of 88.61% and a balanced accuracy of 89.55% on the independent test dataset. Feature importance analysis identifies the second-period grade as the most influential predictor. The findings demonstrate the potential of machine learning-based academic risk prediction as a decision-support approach for educational institutions to identify students who may benefit from timely intervention.

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Submission Files
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Submission Summary

Conference Name
2028 International Conference on Artificial Intelligence in Photonic Systems

Paper ID
135

Paper Title
Automated Student Performance Prediction Using AutoML

Abstract
Early identification of students at academic risk is an important task in Educational Data Mining, as it enables timely academic support and informed educational decision-making. This paper presents an automated student performance prediction framework using machine learning techniques. The proposed approach utilizes the Student Performance Mathematics dataset from the UCI Machine Learning Repository and formulates student performance prediction as a binary classification problem, categorizing students into At Risk and Performing Well groups based on their final mathematics grades. A preprocessing pipeline is developed to handle numerical and categorical attributes using standardization and one-hot encoding. A Gradient Boosting classifier is employed as the baseline model, followed by hyperparameter optimization using RandomizedSearchCV with stratified five-fold cross-validation and balanced accuracy as the optimization metric. The models are evaluated using accuracy, balanced accuracy, precision, recall, F1-score, and confusion matrix analysis, with particular emphasis on identifying students who may require academic support. The baseline model achieves an accuracy of 89.87% and a balanced accuracy of 91.47%, while the optimized model achieves an accuracy of 88.61% and a balanced accuracy of 89.55% on the independent test dataset. Feature importance analysis identifies the second-period grade as the most influential predictor. The findings demonstrate the potential of machine learning-based academic risk prediction as a decision-support approach for educational institutions to identify students who may benefit from timely intervention.

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Submission Files
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APPENDIX_C – SOURCE CODE

A.1 Importing Required Libraries

```
import pandas as pd
import numpy as np
import optuna
import joblib
from sklearn.model_selection import (
    train_test_split,
    StratifiedKFold,
    cross_val_score
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import (
    StandardScaler,
    OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from xgboost import XGBClassifier
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    balanced_accuracy_score,
    confusion_matrix,
    classification_report)
import warnings
warnings.filterwarnings("ignore")
```

A.2 Dataset Loading

```
math_data = pd.read_csv(
    "student-mat.csv",
    sep=";")
portuguese_data = pd.read_csv(
    "student-por.csv",
    sep=";")
print("Mathematics Dataset:", math_data.shape)
print("Portuguese Dataset:", portuguese_data.shape)
print(math_data.head())
print(portuguese_data.head())
```

A.3 Data Preprocessing and Target Creation

```
def prepare_dataset(data, scenario):
    data = data.copy()
```

```
    # Remove duplicate records
    data = data.drop_duplicates()
    # Create binary target variable
    data["Performance"] = (
        data["G3"] >= 10
    ).astype(int)
    # Select features based on prediction scenario
    if scenario == "Early Prediction":
        X = data.drop(
            columns=["G1", "G2", "G3", "Performance"])
    elif scenario == "Grade-Assisted Prediction":
        X = data.drop(
            columns=["G3", "Performance"] )
    else:
        raise ValueError("Invalid prediction scenario")
    y = data["Performance"]
    return X, y
```

A.4 Feature Transformation

```
def create_preprocessor(X):
    numerical_features = X.select_dtypes(
        include=["int64", "float64"]
    ).columns.tolist()
    categorical_features = X.select_dtypes(
        include=["object"]
    ).columns.tolist()
    numerical_pipeline = Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler", StandardScaler()) ])
    categorical_pipeline = Pipeline([
        ("imputer", SimpleImputer(strategy="most_frequent")),
        ("encoder", OneHotEncoder(
            handle_unknown="ignore"))])
    preprocessor = ColumnTransformer([
        ("numerical", numerical_pipeline, numerical_features),
        ("categorical", categorical_pipeline, categorical_features) ])
    return preprocessor
```

A.5 Machine Learning Model Selection

```
def get_model(model_name, trial=None):
    if model_name == "Logistic Regression":
        return LogisticRegression(
            C=trial.suggest_float("C", 0.01, 10.0, log=True)
            if trial else 1.0,
            class_weight="balanced",
            max_iter=2000)
    elif model_name == "Decision Tree":
        return DecisionTreeClassifier(
```

```
            max_depth=trial.suggest_int("max_depth", 3, 20)
            if trial else None,
            class_weight="balanced",
            random_state=42 )
    elif model_name == "Random Forest":
        return RandomForestClassifier(
            n_estimators=trial.suggest_int(
                "n_estimators", 100, 500
            ) if trial else 200,
            max_depth=trial.suggest_int(
                "max_depth", 3, 20
            ) if trial else None,
            class_weight="balanced",
            random_state=42)
    elif model_name == "SVM":
        return SVC(
            C=trial.suggest_float("C", 0.1, 10.0, log=True)
            if trial else 1.0,
            kernel="rbf",
            class_weight="balanced",
            probability=True)
    elif model_name == "XGBoost":
        return XGBClassifier(
            n_estimators=trial.suggest_int(
                "n_estimators", 100, 500
            ) if trial else 200,
            max_depth=trial.suggest_int(
                "max_depth", 3, 10
            ) if trial else 5,
            learning_rate=trial.suggest_float(
                "learning_rate", 0.01, 0.3
            ) if trial else 0.1,
            eval_metric="logloss",
            random_state=42)
    else:
        raise ValueError("Unknown model")
```

A.6 Optuna Hyperparameter Optimization

```
def optimize_model(model_name, X_train, y_train):
    preprocessor = create_preprocessor(X_train)
    cv = StratifiedKFold(
        n_splits=5,
        shuffle=True,
        random_state=42)
    def objective(trial):
        classifier = get_model(model_name, trial)
        pipeline = Pipeline([
```

```
            ("preprocessor", preprocessor),
            ("classifier", classifier) ])
        scores = cross_val_score(
            pipeline,
            X_train,
            y_train,
            cv=cv,
            scoring="f1_macro",
            n_jobs=-1 )
        return scores.mean()
    study = optuna.create_study(
        direction="maximize",
        sampler=optuna.samplers.TPESampler(seed=42))
    study.optimize(
        objective,
        n_trials=50)
    return study.best_params, study.best_value
```

A.7 Model Training and Evaluation

```
def train_and_evaluate(
    model_name,
    X_train,
    X_test,
    y_train,
    y_test,
    best_params):
    preprocessor = create_preprocessor(X_train)
    if model_name == "XGBoost":
        classifier = XGBClassifier(
            **best_params,
            eval_metric="logloss",
            random_state=42 )
    elif model_name == "Random Forest":
        classifier = RandomForestClassifier(
            **best_params,
            class_weight="balanced",
            random_state=42)
    elif model_name == "Decision Tree":
        classifier = DecisionTreeClassifier(
            **best_params,
            class_weight="balanced",
            random_state=42 )
    elif model_name == "Logistic Regression":
        classifier = LogisticRegression(
            **best_params,
            class_weight="balanced",
            max_iter=2000)
```

```
    else:
        classifier = SVC(
            **best_params,
            kernel="rbf",
            class_weight="balanced",
            probability=True )

    pipeline = Pipeline([
        ("preprocessor", preprocessor),
        ("classifier", classifier) ])
    pipeline.fit(X_train, y_train)
    predictions = pipeline.predict(X_test)
    results = {
        "Accuracy": accuracy_score(y_test, predictions),
        "Precision": precision_score(
            y_test, predictions, zero_division=0 ),
        "Recall": recall_score(
            y_test, predictions, zero_division=0),
        "F1 Score": f1_score(
            y_test, predictions, zero_division=0),
        "Macro-F1": f1_score(
            y_test, predictions,
            average="macro",
            zero_division=0 ),
        "Balanced Accuracy": balanced_accuracy_score(
            y_test, predictions)  }
    print("Model:", model_name)
    print(results)
    print(confusion_matrix(y_test, predictions))
    return pipeline, results
```

A.8 Model Serialization Using Joblib

```
# Save the trained model pipeline
joblib.dump(
    best_model,
    "student_performance_model.pkl")
# Load the saved model
loaded_model = joblib.load(
    "student_performance_model.pkl")
print("Model loaded successfully")
```

A.9 Student Performance Prediction

```
def predict_student_performance(model, student_data):
    prediction = model.predict(student_data)

    probabilities = model.predict_proba(student_data)
    if prediction[0] == 1:
        result = "Pass"
```

```
    else:
        result = "Fail"
    confidence = max(probabilities[0]) * 100
    return result, confidence
```

A.10 Streamlit Dashboard

```
import streamlit as st
import pandas as pd
import joblib
st.set_page_config(
    page_title="Student Performance Prediction",
    layout="wide")
st.title("Automated Student Performance Prediction")
st.write(
    "An AutoML-based system for predicting student "
    "academic performance.")
subject = st.selectbox(
    "Select Subject",
    ["Mathematics", "Portuguese"])
scenario = st.selectbox(
    "Select Prediction Scenario",
    ["Early Prediction", "Grade-Assisted Prediction"])
st.subheader("Student Information")
student_file = st.file_uploader(
    "Upload student input data",
    type=["csv"])
if student_file is not None:
    student_data = pd.read_csv(student_file)
    st.write("Student Input Data")
    st.dataframe(student_data)
    if st.button("Predict Performance"):
        model_path = (
            f"models/{subject.lower()}_"
            f"{scenario.lower().replace(' ', '_')}.pkl" )
        model = joblib.load(model_path)
        predictions = model.predict(student_data)
        probabilities = model.predict_proba(student_data)
        output = student_data.copy()
        output["Prediction"] = [
            "Pass" if value == 1 else "Fail"
            for value in predictions]
        output["Confidence"] = probabilities.max(axis=1) * 100
        st.subheader("Prediction Results")
        st.dataframe(output)
        st.success("Prediction completed successfully.")
```

A.11 Execution of the Prediction Framework

```
# Dataset configuration
```

```
datasets = {
    "Mathematics": math_data,
    "Portuguese": portuguese_data}
scenarios = [
    "Early Prediction",
    "Grade-Assisted Prediction"]
models = [
    "Logistic Regression",
    "Decision Tree",
    "Random Forest",
    "SVM",
    "XGBoost"]
random_seeds = [42, 43, 44]
# Iterate through datasets and scenarios
for subject, dataset in datasets.items():
    for scenario in scenarios:
        print("\nSubject:", subject)
        print("Scenario:", scenario)
        X, y = prepare_dataset(dataset, scenario)
        X_train, X_test, y_train, y_test = train_test_split(
            X,
            y,
            test_size=0.20,
            stratify=y,
            random_state=42 )
        for model_name in models:
            print("\nEvaluating:", model_name)
            best_params, best_score = optimize_model(
                model_name,
                X_train,
                y_train)
            best_model, metrics = train_and_evaluate(
                model_name,
                X_train,
                X_test,
                y_train,
                y_test,
                best_params)
            print("Best CV Macro-F1:", best_score)
```

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